QP CODE: 24900145

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR

EXAMINATION NOVEMBER 2024

First Semester

Discipline Specific Core Course - MG1DSCIAM100 -

INTERACTIVE ROBOTIC SYSTEMS

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

[Learning Domain][CO No(s)]

Part A

Multiple Choice Questions

Answer all questions.Each question carries 1 mark

1 What is the microcontroller used in the Arduino Board

a) ATmega328p b) ATmega2560

c) ATmega32U4 d) ATmega8 Ans: A [K] [1]

2 Material used for making of LDR is_____

a) Cadmium sulphide b) Silicon

c) Silver d) Copper Ans: A [K] [2]

3 Which type of wheel mechanism allows a robot to move in any direction without the need to change its orientation?

a) Differential drive wheels b) Ackermann steering wheels

c) Mecanum or omni-wheels d) Skid-steer wheels Ans: C [K] [3]

4 What type of signal does the analogWrite() function output?

a) Pulse Code Modulated Signal b) Frequency Modulated Signal

c) Pulse Width Modulated Signal d) Pulse Amplitude Modulated Signal

Ans: C [K] [1]

5 The basic principle behind the operation of an ultrasonic sensor is

a) Infrared reflection b) Radio wave transmission

c) Sound wave reflection d) Light diffraction Ans: C [U] [2]

6 In a RADAR system utilizing an ultrasonic sensor, what does the term "echo" refer to?

a) The initial sound wave emitted by the sensor

b) The reflection of sound waves off an object

c) The failure of the sensor to detect an object

d) The signal sent to the servo motor Ans: B [K] [3]

7 Which Function is called repeatedly in an Arduino program?

a) void setup() b) void loop()

c) main() d) init() Ans: B [U] [1]

8 The principle of operation of MQ 2 smoke sensor is the change of _____ in the presence of gas

a) voltage b) resistance

c) current d) capacitance Ans: B [U] [2]

9 Which model is commonly used to represent self-balancing robots?

a) Simple Pendulum Model b) Inverted Pendulum Model

c) Circular Motion Model d) Rotational Model

Ans: B [U] [3]

10 Identify the communication protocol used by the MPU6050 to communicate with microcontrollers.

a) UART b) SPI

c) I2C d) CAN

Ans: C [K] [3]

(10 × 1 = 10)

Part B

Short Answer Questions

Answer 4 questions.Each question carries 5 marks

11 Explain how you would Troubleshoot an Arduino program that is not executing the void loop () function as expected. What steps would you take to identify and fix the issue. [A] [1]

Three or more troubleshoot method - 5 mark, Any two method - 3mark

Any one method - 1mark

Ans:

Verify Board and Port Selection : Check the correct board and COM port are selected in the Arduino IDE

Check for Syntax Errors : Verify the code compiles successfully without errors

Verify setup() Completes Execution: Ensure there is no infinite loop like while() or for() in setup() that prevents it from exiting.

Debug Using Serial Communication: Insert Serial.println() statements at key points in loop() to check if the function is being executed.

12 Why servo motor is more suitable for robotic arm rather than DC motor? [U] [2]

Three or more advantage - 5 mark Any two method - 3 Any one method - 1

Ans:

- Precise position control.
- Built-in feedback mechanism.
- High torque at low speeds.
- Programmable and smooth motion.
- Compact and efficient design

13 Explain the significance of PWM signals in robotic application [U] [1]

Three or more - 5 mark Any two - 3 mark Any one - 1 mark

Ans:

Speed control of motors through varying duty cycle.

Precise position control for servo motors.

Energy-efficient power delivery.

Variable force or torque adjustment.

Smooth motion control for robotic components.

Cost-effective and reliable control method

14 How can an autonomous robotic vehicle navigate and change direction in response to obstacles? [U] [3]

Explanation with sensors and actuators - 5 mark

Points only - 3 mark

Ans:

- Sensors for obstacle detection .
- Path planning algorithms
- Motor control and steering for direction changes.
- Feedback mechanism
- Collision avoidance systems.
- Machine learning for advanced perception and decision-making

15 What is the pourpose of an H bridge circuit ? [U] [2]

- Definition of H-bridge circuit (1 mark).
- Motor direction control (1 mark).
- Speed control using PWM (1 mark).
- Braking feature for precise control (1 mark).
- Interface between microcontroller and motor (1 mark)

16 Explain how obstacle detection with IR sensors can be implemented in a real-world robotic application, such as an autonomous vehicle. [U] [3]

- Explanation of IR sensors and their working principle (1 mark)
- Obstacle detection mechanism using reflection of infrared light (1 mark)
- Integration of IR sensors with the robotic control system (1 mark)
- Real-time obstacle avoidance actions based on sensor input (1 mark)
- Limitations and environmental considerations of IR sensors (1 mark)

Part C

Essay Questions

Answer 2 questions.Each question carries 10 marks

17 Explain how to use the delay() function in Arduino to create between LED state changes.Include an example program and explain how the delay value affects the LED's behavior? [U] [1]

- Explanation of the delay() function in Arduino (3 marks).
- Example program using delay() to blink an LED (4 marks).
- Explanation of how delay value affects LED behavior (3 marks).

18 Write a program to read the values from LDR with proper steps? [A] [2]

- Program to read LDR values (10 marks)

```
int ldrPin = A0; // Pin connected to LDR
int ldrValue = 0; // Variable to store LDR reading
void setup() {
  Serial.begin(9600); // Start serial communication
}
void loop() {
  ldrValue = analogRead(ldrPin); // Read the LDR value
  Serial.println(ldrValue); // Print the value to serial monitor
  delay(1000); // Wait for 1 s
```

19 Discuss the working principles of the MPU6050 sensor. Explain how it measures acceleration and angular velocity [U] [3]

- Introduction to MPU6050 sensor and its components (3 marks)
- Working principle of the gyroscope (3 marks)
- How the MPU6050 measures acceleration (2 marks)
- How the MPU6050 measures angular velocity (2 marks)

20 Develop a program for robot car which stop the movement when an obstacle is detected by the IR sensor [U] [2]

- Explanation of robot car components and setup (4 marks).
- Logic for detecting obstacles using the IR sensor (2 marks).
- Developing the program to stop the car when an obstacle is detected (4 marks).

(2 × 10 = 20)